Ollscoil Chathair Bhaile Átha Cliath
Dublin City University

SEMESTER 2 EXAMINATIONS 2024/2025

MODULE: CSC1018 - Logic

PROGRAMME(S):

CASE	BSc in Computer Applications (Sft.Eng.)
COMSCI	BSc in Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,2

EXAMINERS: Dr. David Sinclair (Internal) (Ext:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1 ***[Total marks: 30]***

Alice, Bob and Cian are spies. Every spy is either a spook or an assassin. No assassin likes noise and all spooks like crowds. Cian dislikes whatever Alice likes and likes whatever Alice dislikes. Alice likes noise and crowds.

1(a) [10 Marks]

Convert the problem statement above into a set of predicate formulae.

1(b) [8 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [12 Marks]

Using *General Resolution*, determine if there a spy who is an assassin but not a spook.

[End Question 1]

QUESTION 2 ***[Total marks: 10]***

[10 Marks]

Without using truth tables, show that

$$(p \to q) \land \neg q \equiv \neg(p \lor q)$$

Clearly indicate which equivalence is used at each step.

[End Question 2]

QUESTION 3 ***[Total marks: 10]***

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$$

Clearly indicate which inference rule is used at each step.

[End Question 3]

QUESTION 4 ***[Total marks: 10]***

[10 Marks]

Using Gentzen system $\mathcal{G}$ prove

$$\neg(X \vee Y) \leftrightarrow \neg X \wedge \neg Y$$

Clearly indicate which inference rule is used at each step.

[End Question 4]

QUESTION 5 ***[Total marks: 10]***

[10 Marks]

Convert the following formula to prenex conjunctive normal form (PCNF)

$$\exists x.\forall y.p(x, y) \rightarrow \forall y.\exists x.p(x, y)$$

[End Question 5]

QUESTION 6 ***[Total marks: 10]***

[10 Marks]

Using Gentzen system $\mathcal{G}$, prove the following predicate formula.

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \forall x.q(x))$$

Clearly indicate which inference rule is used at each step.

[End Question 6]

Section B

QUESTION 7 ***[Total marks: 10]***

[10 Marks]

Write a Prolog relation $split(N, Positive, Negative)$ that is true if all the positive numbers (including 0) in the list N are in the list $Positive$, and all the negative numbers in the list N are in the list $Negative$.

[End Question 7]

QUESTION 8 ***[Total marks: 10]***

[10 Marks]

Write a Prolog predicate `oddsquaresum(X, S)` that is true when `S` is the sum of the square of all the odd numbers in the list `X`. Note that the list `X` can contain any atom, e.g. `[4,oranges,5,2,7,apples,1]`. Hence, `oddsquaresum([4,oranges,5,2,7,apples,1], 75)` is true.

[End Question 8]

QUESTION 9 ***[Total marks: 10]***

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2
$\neg\neg A_1$	A_1	
$A_1 \wedge A_2$	A_1	A_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$

β	β_1	β_2
$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$B_1 \vee B_2$	B_1	B_2
$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \downarrow B_2)$	B_1	B_2
$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$

γ	$\gamma(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U_1' \cup \{\alpha_1, \alpha_2\}}{\vdash U_1' \cup \{\alpha\}} \qquad \frac{\vdash U_1' \cup \{\beta_1\} \qquad \vdash U_2' \cup \{\beta_2\}}{\vdash U_1' \cup U_2' \cup \{\beta\}}$$

$$\frac{\vdash U_1' \cup \{\gamma, \gamma(a)\}}{\vdash U_1' \cup \{\gamma\}} \qquad \frac{\vdash U_1' \cup \{\delta(a)\}}{\vdash U_1' \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2
$\neg\neg A$	A	
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$
$A_1 \vee A_2$	A_1	A_2
$A_1 \rightarrow A_2$	$\neg A_1$	A_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$

β	β_1	β_2
$B_1 \wedge B_2$	B_1	B_2
$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$\neg(B_1 \uparrow B_2)$	B_1	B_2
$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$

γ	$\gamma(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]